

Problem

A series RLC circuit has $R = 80\ \Omega$, $L = 240\text{ mH}$, and $C = 5\text{ mF}$. If the input voltage is $v(t) = 115 \cos 2t$, find the current flowing through the circuit.

Step-by-step solution

Step 1 of 4

The sinusoidal expression of input voltage is, $v(t) = 115 \cos(2t)\text{ V}$

Here,

The magnitude of the voltage v_m is 115 V

The angular frequency ω is 2 rad/s

The phase angle is 0° .

The phasor representation of voltage is,

$$\mathbf{V} = 115 \angle 0^\circ\text{ V}$$

Step 2 of 4

Calculate the impedance of 240 mH inductor.

$$\begin{aligned}\mathbf{Z}_L &= j\omega L \\ &= j(2)(240 \times 10^{-3}) \\ &= j0.48\end{aligned}$$

Calculate the impedance of 5 mF capacitor.

$$\begin{aligned}\mathbf{Z}_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(2)(5 \times 10^{-3})} \\ &= -j100\end{aligned}$$

Step 3 of 4

Determine the equivalent impedance of the series RLC circuit.

$$\begin{aligned}\mathbf{Z}_{in} &= R + \mathbf{Z}_L + \mathbf{Z}_C \\ &= 80 + j0.48 - j100 \\ &= 80 - j99.52 \\ &= 127.69 \angle -51.21^\circ\ \Omega\end{aligned}$$

Step 4 of 4

Calculate the current through the circuit.

$$\begin{aligned}\mathbf{I} &= \frac{\mathbf{V}}{\mathbf{Z}_{\text{in}}} \\ &= \frac{115\angle 0^\circ}{127.69\angle -51.21^\circ} \\ &= 0.9006\angle 51.21^\circ \text{ A} \\ &= 900.6\angle 51.21^\circ \text{ mA}\end{aligned}$$

Convert the current expression from phasor domain to time domain.

$$\begin{aligned}i &= i_m \cos(\omega t + \phi) \\ &= 900.6 \cos(2t + 51.21^\circ) \text{ mA}\end{aligned}$$

Therefore, the current through the series RLC circuit is $\boxed{900.6 \cos(2t + 51.21^\circ) \text{ mA}}$.